ACTIVIDAD GUIADA #1 M7C2 - Piñones Huerta, Lorena

Actividad guiada: Primer acceso a datos con el modelo Auto

1) Creación del proyecto Django

```
PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2> & C:\Users\Lorena
\Documents\Curso Python\M7\Clase 2\m7c2\Scripts\Activate.ps1"
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2> pip install
 django
Collecting django
  Using cached django-6.0.3-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref>=3.9.1 (from django)
  Using cached asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.5.0 (from django)
  Using cached sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Collecting tzdata (from django)
  Using cached tzdata-2025.3-py2.py3-none-any.whl.metadata (1.4 kB)
Using cached django-6.0.3-py3-none-any.whl (8.4 MB)
Using cached asgiref-3.11.1-py3-none-any.whl (24 kB)
Using cached sqlparse-0.5.5-py3-none-any.whl (46 kB)
Using cached tzdata-2025.3-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.11.1 django-6.0.3 sqlparse-0.5.5 tzdata
-2025.3

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2> django-admi
n startproject autos_project
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2>
```

Se crea proyecto nuevo en entorno virtual.

2) Integración con la base de datos

```
settings.py
autos_project > settings.py > ...

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': 'db_clase2m7',
        'USER': 'usuariom7',
        'PASSWORD': '123456',
        'HOST': 'localhost',
        'PORT': '5432'
    }
}

-2025.3

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2> django-admin startproject autos_proj
ect
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2> cd autos_project
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project>
```

Se modifica archivo *settings* con configuración para integrar base de datos correspondiente, con instalación de *psycopg2-binary*.

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3) Creación de la aplicación

```
PROBLEMS    OUTPUT    TERMINAL    ...        powershell - autos_project

(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project> pip install psycopg2-b
inary
Collecting psycopg2-binary
  Using cached psycopg2_binary-2.9.11-cp314-cp314-win_amd64.whl.metadata (5.1 kB)
Using cached psycopg2_binary-2.9.11-cp314-cp314-win_amd64.whl (2.8 MB)
Installing collected packages: psycopg2-binary
Successfully installed psycopg2-binary-2.9.11

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project> python manage.py start
app autos
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project>
```

Tras configurar base de datos e integrarla al proyecto se crea app *autos*.

4) Creación del modelo Auto

```
settings.py    models.py ×
autos_project > autos > models.py > ...
1   from django.db import models
2
3   # Create your models here.
4
5   class Auto(models.Model):
6       marca = models.CharField(max_length=50)
7       año = models.IntegerField()
8
9       def __str__(self):
10          return self.marca
11
```

Se crea modelo Auto para albergar datos de marca y año de autos.

```
TERMINAL    ...    powershell - autos_project

(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_proje
ct> python manage.py makemigrations
Migrations for 'autos':
  autos\migrations\0001_initial.py
    + Create model Auto
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_proje
ct> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, autos, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying autos.0001_initial... OK
  Applying sessions.0001_initial... OK
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_proje
ct> }
```

```
EXPLORER
CLASE 2
  autos_project
    autos
      __pycache__
      migrations
      __init__.py
      admin.py
      apps.py
      models.py
      tests.py
      views.py
    autos_project
      __pycache__
      __init__.py
      asgi.py
      settings.py
      urls.py
      wsgi.py
    manage.py
    IMG PROOF
    m7c2
```

5) Migraciones

Se realizan migraciones para poder reflejar la creación del modelo tanto en VSC como en su base de datos correspondiente en PostgreSQL.

6) Inserción de registros

```
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project> python manage.py she
ll
13 objects imported automatically (use -v 2 for details).

Ctrl click to launch VS Code Native REPL
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win3
2
Type "help", "copyright", "credits" or "license" for more information.
(InteractiveConsole)
>>>
```

0 0 Select Postgres Server UTF-8 CRLF { } Python m7c2 (3.14.2) Go Live Prettier

Query Query History

```
SELECT * FROM public.autos_auto
ORDER BY id ASC
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	id [PK] bigint	marca character varying (50)	año integer
1	1	Toyota	2020
2	2	Ford	2018

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL python - autos_project

```
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
(m7c2) PS C:\Users\Lorena\Documents\Curso Python\M7\Clase 2\autos_project> python manage.py she
ll
13 objects imported automatically (use -v 2 for details).

Ctrl click to launch VS Code Native REPL
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win3
2
Type "help", "copyright", "credits" or "license" for more information.
(InteractiveConsole)
>>> Auto.objects.create(marca="Toyota", año=2020)
<Auto: Toyota>
>>> Auto.objects.create(marca="Ford", año=2018)
<Auto: Ford>
>>>
```

Se ejecuta comando *python manage.py shell* para ejecutar el shell que nos permitirá realizar consultas y crear objetos sin tener que acudir a pgadmin. Tras eso se procede a crear 2 objetos y se verifica en pgadmin.

7) Consultas ORM: all() y get()

```
>>> Auto.objects.all()
<QuerySet [<Auto: Toyota>, <Auto: Ford>]>
>>> Auto.objects.get(id=1)
<Auto: Toyota>
>>>
```

Lo último fue consultar tanto por todos los objetos existentes con *all()* como también probar obtener solo un (1) dato con *get()*.